Fundamentals of Deep Learning and Programming

Lecture 0: Course Overview

Fall 2026

Adjunct Professor: Donghoon Kang

Course Overview

Lecture Time	Fridays, 2 PM – 5 PM
Classroom	Room: L3 301, 302 (3334A, B)
Evaluation Criteria	Attendance (20%) Programming Assignments (10%) Term-project (20%) Mid-term (25%), Final (25%)

- Programming assignments: students should submit the code and 1 ~ 2 page document briefly explaining the code to me by email (chocopie@kist.re.kr)

Course Overview

- If you are unable to attend class due to official travel or conference participation, **please notify me in advance by email (chocopie@kist.re.kr)**.

  Absences without prior notification will result in a deduction of points.

Course Website

`https://kimbabmoowoo.github.io/lecture.html`

Publications   Research   Contact

UST Course Fall Semester 2025

Fundamentals of Deep Learning and Programming

1. **Week 1 (9.05):** Course overview `pdf` and introduction to Python programming `pdf`
2. **Week 2 (9.12):** Useful Python Libraries `pdf` and code `zip`
3. **Week 3 (9.19):** Neural Network Basics (1) `pdf` and code `zip`
4. **Week 4 (9.26):** Neural Network Basics (2) `pdf` and code `zip`
5. **Week 5 (10.10):** Convolutional Neural Network (CNN) (1) `pdf` and code `zip`
6. **Week 6 (10.24):** Midterm exam `pdf` and the solution `pdf`
7. **Week 7 (10.31):** Convolutional Neural Network (CNN) (2) and Introduction to Natural Language Processing (NLP) (0) `pdf`
8. **Week 8 (11.14):** Introduction to Natural Language Processing (NLP)(1) `pdf` and code `zip`
9. **Week 9 (11.21):** Introduction to Natural Language Processing (NLP)(2) & Attention Mechanism `pdf` and code `zip`
10. **Week 10 (11.28):** Transformer (1) `pdf` and code `zip`
11. **Week 11 (12.05):** Transformer (2) `pdf`

    Two papers for final exam preparation: "Attention is all you need" (NeurIPS 2017) `pdf` and "An image is worth 16x16 words" (ICLR 2021) `pdf`
12. **Week 12 (12.12):** Diffusion Models `pdf`
13. **Week 13 (12.19):** Final exam `pdf` and the solution `pdf`

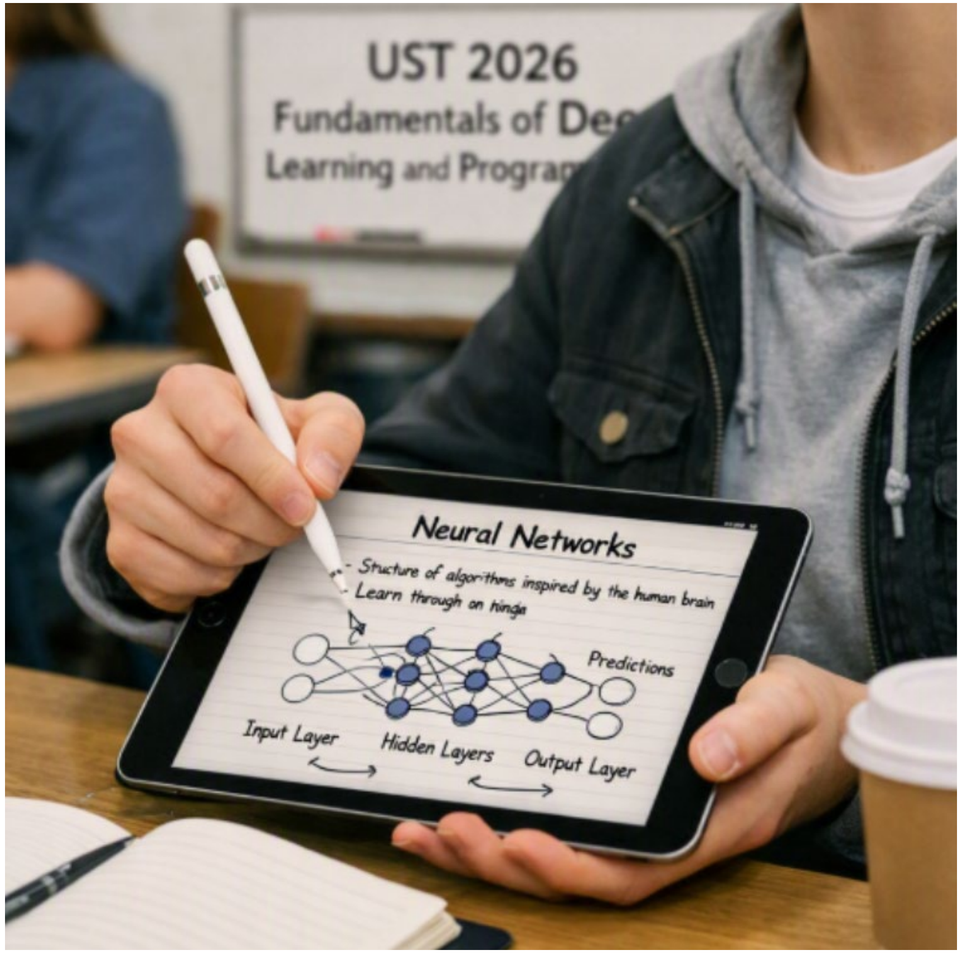
Term Project Reports

- Mr. Suleman Khan `pdf` ("EMGBench: Benchmarking Out-of-Distribution Generalization and Adaptation for Electromyography", NeurIPS 2024) `pdf`
- Mr. JeongMin Heo `pdf` ("NightReID: A Large-Scale Nighttime Person Re-Identification Benchmark", AAAI 2025) `pdf`
- Mr. DongHyeok Min `pdf` ("BEVFormer: Learning Bird's-Eye-View Representation from Multi-Camera Images via Spatiotemporal Transformers", ECCV 2022) `pdf`

Highly Recommended

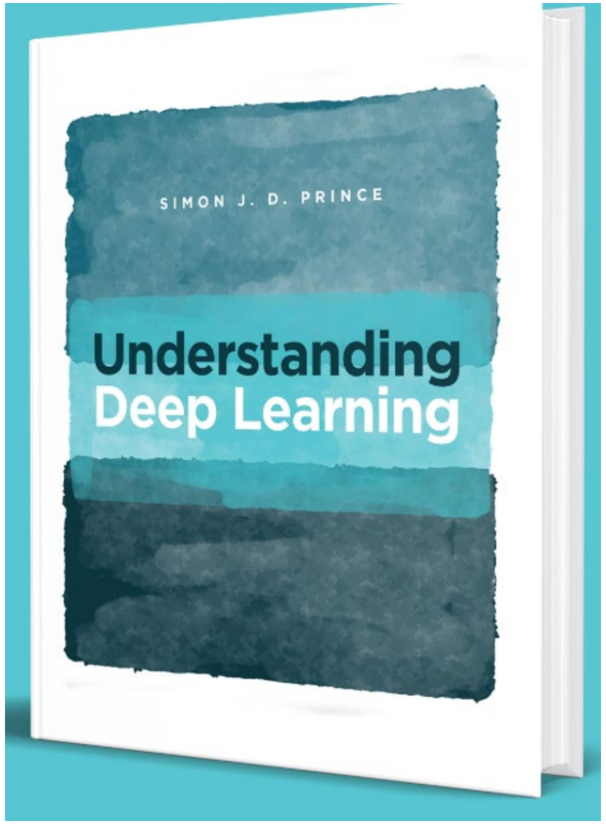
I plan to **upload** the **lecture slides** in PDF format to the course website at around **1:00 p.m. each Friday**.

Students are strongly encouraged to **download the PDF** to their **iPad** or **laptop** before class so that they can view it and **take notes during the lecture.**



Recommended book (pdf is free)

`https://udlbook.github.io/udlbook/`



Understanding Deep Learning
by Simon J.D. Prince | Dec 5, 2023
168

Customer reviews
4.9 out of 5
168 global ratings

5 star	94%
4 star	3%
3 star	1%
2 star	0%
1 star	2%

Download full PDF (29 May 2025)

downloads 523k

2026 Calendar

달력 | 양음력변환 | 전역일계산

오늘 < 2026.09 > 음력 손없는날 기념일

일	월	화	수	목	금	토
30	31	1 통계의 날	2	3	4 지식재산...	5
6	7 백로 사회복지... 푸른하늘...	8	9 숙련기술...	10 9.10 해양... 세계 자살...	11 음8.1	12
13 대한민국 ...	14	15	16	17	18	19 청년의날
20	21 치매극복...	22	23 추분 이산가족...	24	25 음8.15 추석	26
27	28	29	30 개인정보 ...	1	2	3

- **Holiday 9.25 (Fri.): Chuseok**
  - the Korean harvest festival

2026 Calendar

달력 | 양음력변환 | 전역일계산

오늘 2026.10 음력 손없는날 기념일

일	월	화	수	목	금	토
27	28	29	30	1 국군의 날	2 노인의 날	3 개천절
4	5 대체 휴일 세계 한인...	6	7	8 한로 재향 군인...	9 한글날	10 임산부의 날 정신건강...
11 음 9.1	12	13	14	15 체육의 날	16 세계 식량... 부마민주...	17
18	19	20	21 경찰의 날	22	23 상강	24 국제연합일
25 음 9.15 독도의날	26	27 금융의 날	28 교정의 날	29 지방자치...	30	31 회계의 날

- **Holiday 10.09 (Fri.): Hangeul** Day
- **Tentative date for KIST sports festival: 10.16 (Fri.)**

2026 Calendar

달력 | 양음력변환 | 전역일계산

오늘 < 2026.11 > 음력 손없는날 기념일

일	월	화	수	목	금	토
1	2	3 학생 독립...	4 점자의 날	5 소상공인...	6	7 입동
8	9 음 10.1 소방의 날	10	11 농업인의 날 유엔참전... 보행자의 날	12	13	14
15	16	17 순국선열...	18	19 아동학대...	20	21
22 소설 김치의 날	23 음 10.15	24	25	26	27	28
29	30	1	2	3	4	5

2026 Calendar

달력 | 양음력변환 | 전역일계산

오늘 2026.12 음력 손없는날 기념일

일	월	화	수	목	금	토
29	30	1	2	3 소비자의 날	4	5 자원봉사... 무역의 날
6	7 대설	8	9 음 11.1	10	11	12
13	14	15	16	17	18	19
20	21	22 동지	23 음 11.15	24	25 성탄절	26
27 원자력의 날	28	29	30	31	1	2

2026 Calendar

달력 | 양음력변환 | 전역일계산

오늘 ‹ 2026.12 › 음력 손없는날 기념일

일	월	화	수	목	금	토
29	30	1	2	3 소비자의 날	4	5 자원봉사... 무역의 날
6	7 대설	8	9 음 11.1	10	11	12
13	14	15	16	17	18	19
20	21	22 동지	23 음 11.15	24	25 성탄절	26
27 원자력의 날	28	29	30	31	1	2

No-class day

- **Holiday 9.25 (Fri.):** Chuseok
  - the Korean harvest festival
- **Holiday 10.09 (Fri.):** Hangeul Day
- **Tentative date for KIST sports festival: 10.16 (Fri.)**

Arranging separate **supplementary classes** for all 10 students would be **difficult**.

Would it be possible to **extend** each regular class by **40 minutes**, changing the class time from **2:00–5:00 p.m.** to **2:00–5:40 p.m.**?

Schedule

September

Week 1 (9.04)	Course overview Introduction to Python Programming
Week 2 (9.11)	PyTorch Fundamentals
Week 3 (9.18)	Neural Networks Training
9.25	No class: Holiday (Chuseok)

Schedule

October

Week 4 (10.02)	Convolutional Neural Networks (CNNs)
10.09	No class: Holiday (Hangeul Day)
10.16	(Tentative) KIST Sports Festival
Week 5 (10.23)	Mid-term Exam
Week 6 (10.30)	Fundamentals of Object Recognition

Schedule

November

Week 7 (11.06)	Fundamentals of Natural Language Processing
Week 8 (11.13)	Transformer (1): Self-Attention and Encoder
Week 9 (11.20)	Transformer (2): Pretraining and Fine-Tuning
Week 10 (11.27)	Generative Models – Introduction to GANs, VAEs

Schedule

December

Week 11 (12.04)	Diffusion Models
Week 12 (12.11)	Fundamentals of Video Recognition
Week 13 (12.18)	Final Exam

Term Project

Must be performed alone

- Should select a paper published only in CVPR, ICCV, ECCV, or NeurIPS after 2022 (inclusive).
- Should write an 4 ~ 6 page report analyzing the "training code" of the selected paper.

Before Week 7 (10.16)

Should decide which paper to choose. (also possible to change another paper later)

Before Week 15 (12.11)

Should submit an 4 ~ 6 page report as a term-project.